William De Rocco

203-209-1600 williamderocco@gmail.com github.com/wderocco8 Seattle, WA

EXPERIENCE

John Hancock — Technology Rotational Program **Boston, MA**

Associate Software Engineer — Fraud, Waste & Abuse Platform *June 2026 – Present*

- Build and operate a fleet of **Go** microservices on **Kubernetes** that process reimbursement claims for fraud, waste, and abuse detection — a pipeline recovering **eight figures** in improper payments annually.
- Root-caused **560** SLA violations: ~**150** traced to a Unicode encoding mismatch between **Azure Blob Storage** and Go's `net/http` that silently dropped uploads, the rest to a 60/hr throughput ceiling; shipped both fixes with monitoring to validate.
- Designed a structured, **type-safe Go rules engine** replacing a complex third-party (`grules`) engine; validated end-to-end and approved for release, resolving **5** known production defects and cutting rule authoring from days to hours.
- Found and resolved **8+** production defects, including a **race condition** duplicating retries across multiple AKS replicas.

Associate Software Engineer — JH AutoForge *June 2025 – June 2026*

- One of **two founding engineers** on a greenfield AI document pipeline (**.NET**, **SQL**, **Azure OpenAI**) converting 200+ page product specs into reviewable Jira user stories, cutting analyst effort an estimated **60–70%**; now adopted by **4+** teams.
- Architected the retrieval pipeline — OCR extraction, keyword-driven chunking, cross-document contextual mapping, and batched context-scoped LLM calls — to minimize hallucination on inputs far exceeding the model context window.
- Owned the team's database workflow, driving migration from containerized local **mssql** to shared lower-tier environments and establishing migration/rollback conventions enforced pre-merge.

Rhapsody Health **Boston, MA**

Full Stack Software Engineering Intern *May 2024 – Aug 2024*

- Established a **Docker** container-driven workflow, standardizing local development and simplifying the **Bitbucket** CI/CD pipeline.
- Built a debounced keyword-based graph search over the component canvas, eliminating input lag during filtering.
- Collaborated with an internationally distributed team using **Jira** and **FigJam** for **project planning** and **design** sprints.

Boston University — Department of Computer Science **Boston, MA**

Undergraduate Teaching Assistant *Jan 2022 – May 2025*

- Tutored **450+** students on algorithms, data analytics, and web development, and configured **Gradescope** test automation to grade **1200+** code samples per semester.

PROJECTS

backtab | *TypeScript, React, Plasmo, Chrome MV3* **2025 – Present**

- Chrome extension reconstructing a browsing session as a navigable **directed graph**, letting users re-enter abandoned branches that linear back-button history discards.
- Reconciles `chrome.webNavigation` with an in-page `window.navigation` tracker to capture same-document SPA route changes `onCommitted` never fires for, filtering both streams so one click yields one node.

Expense Report Generator | *Next.js, AWS (S3, SQS, Lambda, Textract), PostgreSQL* **2026 – Present**

- Generates company-specific expense reports from receipt photos against **user-defined Excel schemas**, built for a user whose employer mandates a fixed format off-the-shelf tools cannot produce.
- Iterated through four architectures — synchronous, fire-and-forget, **event-driven S3/SQS/Lambda**, then two-phase **Textract** OCR plus LLM extraction — resolving Vercel serverless execution semantics and upstream rate limiting.

WhyPhi | *Technical Lead & VP of Technology* **Jan 2024 – May 2025**

- Built and operated an application and attendance platform for a **126-member** organization, processing **314** rush applicants on a layered **AWS Chalice** backend; retired three paid third-party services, cutting recurring tooling costs ~**60%** annually.
- Led the project for over a year — past the end of my elected VP term — then recruited and onboarded a successor team and handed off ownership before graduating.
- Led a team of **12** to plan and execute a **40+** person community-wide **HackerRank** coding competition.

SKILLS

Languages: Go, TypeScript, Python, SQL, C#, Java, Bash/Unix, YAML, HTML/CSS
Infrastructure & Tools: Kubernetes, Docker, Azure, AWS (Lambda, S3, SQS, Textract), PostgreSQL, GitHub Actions, GCP
Frameworks: .NET, React.js, Next.js, Node.js, .NET, Django, Drizzle ORM

EDUCATION

Boston University **Boston, MA**

B.A. in Computer Science, Minor in Economics — GPA 3.86/4.00 *May 2025*